

2017 年真题参考答案

一、选择题

(1) A. (2) B. (3) D. (4) C. (5) D. (6) C. (7) B. (8) B.

二、填空题

(9) $y = x + 2$. (10) $-\frac{1}{8}$. (11) 1. (12) xye^y . (13) $-\ln(\cos 1)$. (14) -1 .

三、解答题

(15) $\frac{2}{3}$.

(16) $\left. \frac{dy}{dx} \right|_{x=0} = f'_1(1,1)$. $\left. \frac{d^2y}{dx^2} \right|_{x=0} = f'_1(1,1) + f''_{11}(1,1) - f'_2(1,1)$.

(17) $\frac{1}{4}$.

(18) 极大值: $y(1) = 1$. 极小值: $y(-1) = 0$.

(19) 证明略.

(20) $\frac{5\pi}{4}$.

(21) $\ln(x^2 + y^2) + 2\arctan \frac{y}{x} = 0$, $x \in \left(0, \frac{3}{2}\right)$.

(22) (I) 证明略.

(II) $k(1, 2, -1)^T + (1, 1, 1)^T$ 为线性方程组 $A\mathbf{x} = \boldsymbol{\beta}$ 的通解, 其中 k 为任意常数.

(23) $a = 2$, $\boldsymbol{Q} = \begin{pmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \\ 0 & -\frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \end{pmatrix}$, 且 $\boldsymbol{Q}^T \boldsymbol{A} \boldsymbol{Q} = \begin{pmatrix} 6 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, 即 f 在正交变换 $\mathbf{x} = \boldsymbol{Q}\mathbf{y}$ 下的标准

形为 $f = 6y_1^2 - 3y_2^2$.